

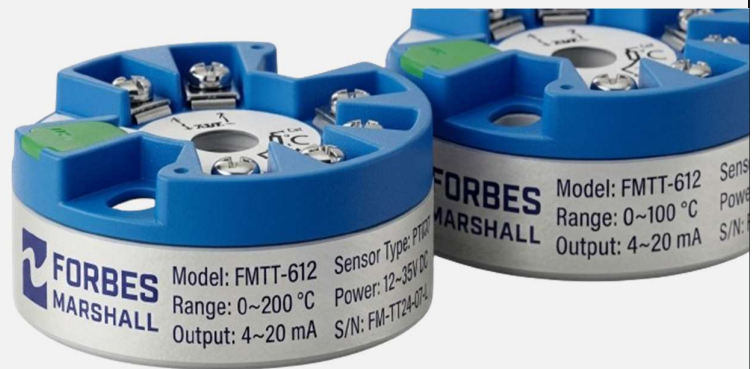
FMTT-012 Head Mount Smart Temperature Transmitter

What is a FMTT-012?

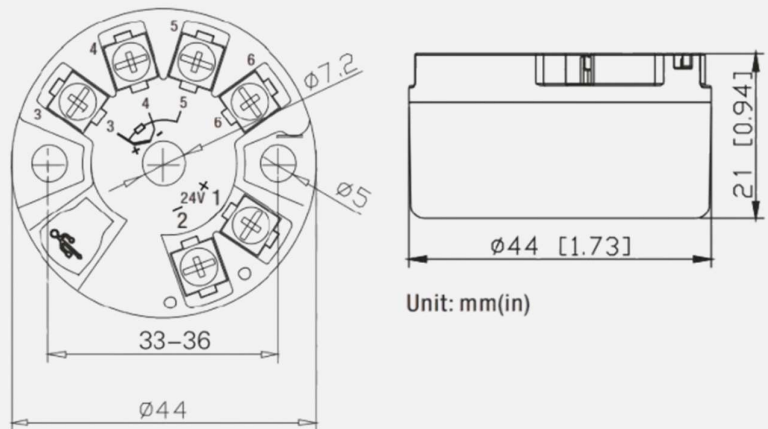
The TT-210 is compact, high-accuracy head-mount smart temperature transmitters designed for industrial temperature measurement applications. Built for reliability and flexibility, these transmitters support a wide range of RTD and Thermocouple inputs and can be configured easily via PC software or Android smartphone.

How it works?

The FMTT-012 converts low-level RTD and thermocouple signals into a stable 4–20 mA output directly at the sensor head, minimizing signal loss, electrical noise, and wiring errors. Advanced signal conditioning ensures accurate linearization, while built-in cold junction compensation continuously corrects thermocouple reference errors. Surge and reverse-polarity protection safeguard the transmitter in harsh industrial environments. Head-mount installation reduces wiring length and improves response time. PC and Android-based configuration enables quick setup, re-ranging, and commissioning without removing the transmitter, ensuring reliable, repeatable temperature measurement throughout the product lifecycle.



Dimension



Special Features

- Universal input support: RTD (Pt100) and Thermocouples (K, J, S, R, N)
- TT-210: 2-wire 4–20 mA output
- High accuracy: $\pm 0.1\%$ F.S. for RTD, $\pm 0.2\%$ F.S. for Thermocouple
- Built-in cold junction compensation
- Configurable via PC software or Android smartphone
- USB configuration cable draws power directly from PC (no separate 24 VDC required)
- Surge protection and reverse polarity protection
- Compact head-mount design, suitable for Form-B connection heads

Industries & Applications

- Chemical & Petrochemical Plants
- Pharmaceutical & Biotechnology Facilities
- Power Generation & Utilities
- Oil & Gas (Upstream, Midstream & Downstream)
- Water & Wastewater Treatment
- Food & Beverage Processing
- Pulp & Paper Industry
- Metals, Cement & Mining
- HVAC & Building Automation
- OEM Machinery & Packaged Skids
- Automation, PLC & DCS Systems
- Refineries & Industrial Utilities
- Renewable Energy Systems

General Specifications – FM TT-012

Model	: FM TT-012
Sensor Type	: RTD: Pt100
	: Thermocouple: K, J, S, R, N
Cold Junction Compensation Range (TC)	: -40 °C to +80 °C
Cold Junction Compensation Accuracy (TC)	: ±1 °C
Accuracy (Ambient 20 °C)	: RTD: ±0.1% of Full Scale
	: TC: ±0.2% of Full Scale
Temperature Drift	: ±0.01% F.S / °C
Response Time	: 1 ms to 90% of final output
Output Signal	: 4–20 mA (2-wire)
Sensor Break Output Current	: 21 mA
Power Supply	: 12–35 VDC
Load Resistance	: $RL \leq (U_e - 12) / 0.021$
Input / Output Isolation	: Non-isolated
Input / Output Impedance	: Non-isolated
Ambient Temperature	: -40 °C to +80 °C
Reference EMC Standard	: IEC 61326-1
Mounting Type	: Head-mount, Form-B compatible
Mounting Screws	: M4 × 2 NO'S
Measuring Range	: RTD: Pt100: -200~600°C, TC: B: 400~1800°C, J/K: -100~1372°C, R/S: -50~1760°C



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Plot No. A-19/2 & T-4/2, I.D.A Nacharam,
Hyderabad - 500 076
Tel: +91-40-27179223, 27171909, 27170732
CIN: U241111TG1991PTC013030

B-85, Phase II, Chakan Indl Area
Sawardari, Chakan, Tal. Khed Dist.

A-34/35, MIDC H Block, Pimpri, Pune-411 018. INDIA.
Tel: 91(0)20-27442020, 39851199
Tel: 91(0)2135-393400 CIN No.: U28996PN1985PTC037806

Email: sales-gauges@forbesmarshall.com, ccmidc@forbesmarshall.com www.forbesmarshall.com

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